**University of Science and Technology of Oran – Mohamed Boudiaf**
Faculty of Mathematics and Informatics
Department of Computer Science

# TP 1 : Hash Tables

Direct access, hash sets and hash maps

Mohammed Said BELAID
said.belaid@univ-usto.dz

20/10/2026

**Objectives**

- Implement a simple hash table.
- Use hash tables in practical examples.
- Replace an $O(n^2)$ search by an $O(n)$ one.

## Exercise 1: Anagrams

Two words are **anagrams** if they contain the same letters, the same number of times.

```
isAnagram("listen", "silent")  ->  true
isAnagram("hello", "world")    ->  false
```

1. **Naive solution.** For each letter of `a`, find an identical letter of `b` not used yet. Write `isAnagram(a, b)`. What is its complexity?
2. **Direct access.** Count the letters of each word in an array of 26 counters (index `c - 'a'`), then compare the arrays. What is the complexity?
3. We now accept any ASCII character (codes 0 to 127). How large must the direct-access array be? Instead, implement a small hash table of 32 buckets with $h(c) = c \bmod 32$ and chaining, with the operations `increment(c)` and `get(c)`. Rewrite `isAnagram` with it.
4. We now count the **words** of a text. Why is direct access no longer possible? Write `wordCounts(words)` with a hash map.

## Exercise 2: Find the duplicate

Rewrite `hasDuplicate(a)` from TP0 with a hash set. Compare it with your TP0 versions (time and memory).

## Exercise 3: Fill a knapsack with two objects

A knapsack has a capacity $C$, and object $i$ has a weight `w[i]`. Find **two** objects that fill the knapsack **exactly**: `w[i] + w[j] = C`.

```
twoObjects([3, 8, 5, 11, 2], 13)  ->  1 and 2
twoObjects([5, 2, 9], 10)         ->  none
twoObjects([5, 2, 5], 10)         ->  0 and 2
```

1. **Naive solution.** Test every pair. What is its complexity?

2. **A better solution.** How can we do better? Write it and give its complexity.

## Exercise 4: A phone book

We want a phone book where a contact can be found by name **and** by phone number, both in $O(1)$ on average. Implement `add(name, phone)`, `findByName(name)`, `findByPhone(phone)` and `remove(name)`.